

REST API SPECIFICATION — TRACE FINDERS

API Version: v1.0.0

Base URL (Vercel Cloud): https://criminalnetworkanalysis.vercel.app/api

Base URL (Local Node): http://localhost:8000/api

Content-Type: application/json

API Endpoints Summary

Method	Path	Description	Query Parameters
GET	/overview	Case overview statistics & activity stream	case_id, person_id
GET	/cases	Registered case list & suspects	None
GET	/persons/{person_id}	Detailed single person profile	Path: person_id
GET	/fusion	Multi-hop evidence chain & XAI assessment	person_id
GET	/graph	Vis.js Knowledge Graph nodes & edges	case_id, person_id
GET	/communications	CDR telecom history & contact tree	person_id
GET	/financial	Financial ledger transactions & Hawala triggers	person_id
GET	/blockchain	EVM/UTXO wallet transactions	person_id
GET	/osint	Public-source crawled intelligence records	person_id
GET	/dvr	CCTV camera inventory & 24h event timeline	person_id
GET	/timeline	Person investigative chronology events	person_id
GET	/evidence/{id}	Single evidence record details & SHA-256 hash	Path: id
GET	/reports/generate	Render person-scoped HTML dossier	case_id, person_id
GET	/search	Global entity & evidence search query	q

Endpoint Specifications

1. GET /api/overview

Retrieves operational summary metrics, recent investigation activity stream, and AI leads.

Query Parameters

- case_id (string, default: "TRX-2026-017")
- person_id (string, default: "P-001")

Response (200 OK)

```
{
  "status": "success",
  "case_summary": {
    "case_id": "TRX-2026-017",
    "title": "OPERATION NEXUS",
    "primary_subject": "Arjun Sharma",
    "evidence_count": 24,
    "relationships_count": 7,
    "communications_count": 421,
    "financial_count": 63,
    "osint_count": 42,
    "blockchain_count": 18,
    "cctv_count": 3
  },
  "investigation_activity": [
    {
      "time": "09:42",
      "event": "Communication activity logged for Arjun Sharma (+91 98765 1201)",
      "domain": "CDR"
    }
  ]
}
```

2. GET /api/fusion

Returns the person-scoped 5-step cross-domain multi-hop evidence chain and XAI rationale.

Query Parameters

- person_id (string, required, e.g. "P-001")

Response (200 OK)

```
{
  "person_id": "P-001",
  "person_name": "Arjun Sharma",
  "role": "Primary Subject",
  "multi_hop_chain": [
    {
      "step": 1,
      "domain": "PROFILE",
      "label": "Arjun Sharma Profile Identified",
      "detail": "Phone: +91 98765 1201 | City: Pune",
      "badge": "EV-PROF-001"
    }
  ],
  "explainable_ai": {
    "title": "MULTI-HOP INTELLIGENCE CORRELATION FOR ARJUN SHARMA",
    "confidence_score": "88%",
    "action_status": "VERIFIED CORRELATION",
    "what": "High cross-domain evidence link score observed.",
    "why": "Matching timestamps between CDR tower log and ANPR camera.",
    "alternative_explanation": "Coincidental proximity during commercial hours.",
    "supporting_evidence": ["EV-COM-001", "EV-CCTV-031"]
  }
}
```

3. GET /api/graph

Returns nodes and edges formatted for interactive Vis.js graph rendering.

Query Parameters

- case_id (string)
- person_id (string)

Response (200 OK)

```
{
  "header_stats": {
    "subject_name": "Arjun Sharma",
    "entities_count": 8,
    "relationships_count": 7,
    "evidence_links_count": 14
  },
  "nodes": [
    {
      "id": "P-001",
      "label": "Arjun Sharma",
      "type": "PERSON",
      "role": "Primary Subject",
      "tree_level": 1
    }
  ],
  "edges": [
    {
      "id": "EDGE-001",
      "source": "P-001",
      "target": "PHONE-P-001",
      "relation": "USES_PHONE",
      "evidence_id": "EV-COM-001"
    }
  ]
}
```

4. GET /api/dvr

Returns CCTV camera inventory (all 12 cameras) and 24-hour event markers.

Response (200 OK)

```
{
  "camera_inventory": [
    {
      "id": "CAM-01",
      "name": "TechDesk Gate 1",
      "location": "Viman Nagar, Pune",
      "camera_type": "Fixed CCTV",
      "status": "Active Recording",
      "recording_window": "00:00 - 23:59",
      "events_count": 14,
      "evidence_links": 6
    }
  ]
}
```